Reteach lessons

One lesson for each question on the four weekly checks. Teach the lesson for the question the PLC picked (the item guide names it). Project the lesson page for the idea and the worked example; students do Your turn and the practice on whiteboards. The page after each lesson has the answers. About 8 to 10 minutes.

Adding and subtracting expressions

Unit 9 · Algebraic Expressions

- **Q1** Adding expressions: match up like terms
- **Q2** Adding decimal terms: add the numbers in front
- **Q3** The missing expression: undo it with subtraction
- **Q4** Subtracting a group: the minus reaches every term
- **Q5** From a situation: write it whole, then simplify

Equivalent expressions

Unit 9 · Algebraic Expressions

- **Q1** Distributing a fraction: multiply every term
- **Q2** Spot the error: multiply both terms, then stop
- **Q3** Distributing a decimal: same rule, careful multiplying
- **Q4** Equivalent or not: simplify both, then compare
- **Q5** A negative outside the parentheses: flip every sign inside

Two-step equations

Unit 10 · Equations and Inequalities

- **Q1** Fee plus rate: write it, then undo in reverse order
- **Q2** Equations with parentheses: undo the outside first
- **Q3** Groups: when the change happens to each one
- **Q4** Fraction equations: add, then use the reciprocal
- **Q5** Two-part problems: solve, then answer the real question

One-step inequalities

Unit 10 · Equations and Inequalities

- **Q1** Reading a graph: the circle and the arrow
- **Q2** Adding or subtracting: the symbol stays put
- **Q3** Dividing by a negative flips the symbol
- **Q4** Words to an inequality: product, sum, at least, at most
- **Q5** Matching a graph: solve every choice, check three things

Adding expressions: match up like terms

Only like terms combine: m-terms with m-terms, numbers with numbers. Every term keeps its own sign.

WORKED EXAMPLE

Simplify:

$$\left(4 + \frac{2}{3}m\right) + \left(\frac{5}{6}m - 7\right)$$

1. Pair the like terms: $\frac{2}{3}m$ with $\frac{5}{6}m$, and 4 with −7.
2. m-terms: $\frac{2}{3} + \frac{5}{6} = \frac{4}{6} + \frac{5}{6} = \frac{9}{6} = \frac{3}{2}$, so $\frac{3}{2}m$.
3. Numbers: $4 + (-7) = -3$.

ANSWER $\frac{3}{2}m - 3$

WATCH FOR

Count both m-terms, not just one.

Keep each sign: the −7 stays negative.

A number never joins an m-term: $\frac{3}{2}m - 3$ is finished.

YOUR TURN

Simplify:

$$\left(5 + \frac{1}{4}k\right) + \left(\frac{3}{8}k - 2\right)$$

PRACTICE · ONE CHANGE EACH TIME

1 $\left(\frac{2}{5}y + 3\right) + \left(\frac{1}{5}y - 8\right)$

same denominators

2 $\left(\frac{1}{2}n - 6\right) + \left(\frac{1}{3}n + 6\right)$

the numbers cancel

3 $\left(\frac{3}{4}w + 1\right) + \left(2 - \frac{1}{2}w\right)$

a w-term is subtracted

Adding expressions: match up like terms

YOUR TURN

Simplify:

$$\left(5 + \frac{1}{4}k\right) + \left(\frac{3}{8}k - 2\right)$$

1. k-terms: $\frac{1}{4} + \frac{3}{8} = \frac{2}{8} + \frac{3}{8} = \frac{5}{8}$.
2. Numbers: $5 + (-2) = 3$.

$\frac{5}{8}k + 3$

PRACTICE

1 $\left(\frac{2}{5}y + 3\right) + \left(\frac{1}{5}y - 8\right)$

$\frac{3}{5}y - 5$

2 $\left(\frac{1}{2}n - 6\right) + \left(\frac{1}{3}n + 6\right)$

$\frac{5}{6}n$

3 $\left(\frac{3}{4}w + 1\right) + \left(2 - \frac{1}{2}w\right)$

$\frac{1}{4}w + 3$

Adding decimal terms: add the numbers in front

$0.4x + 1.6x$ means 0.4 of x plus 1.6 of x, which is $2x$. Add the coefficients; the x stays the same.

WORKED EXAMPLE

Simplify:

$$(1.3x + 4) + (2.2x - 5.5)$$

1. x-terms: $1.3x + 2.2x = 3.5x$.
2. Numbers: $4 + (-5.5) = -1.5$.

ANSWER $3.5x - 1.5$

WATCH FOR

Add the coefficients, don't multiply them: $0.4 \times 1.6 = 0.64$ has nothing to do with it.

An x-term and a number stay separate: $2x - 0.5$ does not become $1.5x$.

Check the sign of the number: $2 + (-2.5) = -0.5$.

YOUR TURN

Simplify:

$$(0.6x + 3) + (2.9x - 4.2)$$

PRACTICE · ONE CHANGE EACH TIME

1 $(4.5y - 1) + (0.5y + 2.5)$

the coefficients make a whole number

2 $(2.75a + 0.5) + (1.25a - 3)$

hundredths

3 $(-1.5c + 6) + (3.2c - 6.4)$

a negative coefficient

Adding decimal terms: add the numbers in front

YOUR TURN

Simplify:

$(0.6x + 3) + (2.9x - 4.2)$

1. x-terms: $0.6 + 2.9 = 3.5$.
2. Numbers: $3 + (-4.2) = -1.2$.

$3.5x - 1.2$

PRACTICE

1 $(4.5y - 1) + (0.5y + 2.5)$

$5y + 1.5$

2 $(2.75a + 0.5) + (1.25a - 3)$

$4a - 2.5$

3 $(-1.5c + 6) + (3.2c - 6.4)$

$1.7c - 0.4$

The missing expression: undo it with subtraction

If $A - ? = B$, then $? = A - B$, just like $10 - ? = 7$ means $? = 10 - 7$. Subtracting a negative adds.

WORKED EXAMPLE

What expression makes it true?

$$\left(\frac{1}{2}x + 4\right) - (__) = -\frac{1}{4}x - 3$$

1. $? = \left(\frac{1}{2}x + 4\right) - \left(-\frac{1}{4}x - 3\right)$.
2. x-terms: $\frac{1}{2} - \left(-\frac{1}{4}\right) = \frac{2}{4} + \frac{1}{4} = \frac{3}{4}$.
3. Numbers: $4 - (-3) = 7$.
4. Check: $\left(\frac{1}{2}x + 4\right) - \left(\frac{3}{4}x + 7\right) = -\frac{1}{4}x - 3$.

ANSWER $\frac{3}{4}x + 7$

WATCH FOR

Two minuses make a plus: $\frac{2}{3} - \left(-\frac{3}{8}\right) = \frac{2}{3} + \frac{3}{8}$.

$6 - (-2)$ is 8, not 4.

Start with the left expression: ? = left − right. Right − left flips every sign.

YOUR TURN

$$\left(\frac{2}{3}x + 1\right) - (__) = -\frac{1}{6}x - 5$$

PRACTICE · ONE CHANGE EACH TIME

1 $(5x + 9) - (__) = 2x + 4$

whole numbers: see the pattern

2 $\left(\frac{3}{5}x - 2\right) - (__) = \frac{1}{5}x - 7$

the result is more negative

3 $\left(\frac{1}{2}x + 3\right) + (__) = \frac{5}{4}x - 1$

a plus sign: now ? = right side − left part

The missing expression: undo it with subtraction

YOUR TURN

$$\left(\frac{2}{3}x + 1\right) - (__) = -\frac{1}{6}x - 5$$

1. x-terms: $\frac{2}{3} - \left(-\frac{1}{6}\right) = \frac{4}{6} + \frac{1}{6} = \frac{5}{6}$.
2. Numbers: $1 - (-5) = 6$.

$\frac{5}{6}x + 6$

PRACTICE

1 $(5x + 9) - (__) = 2x + 4$

$3x + 5$

2 $\left(\frac{3}{5}x - 2\right) - (__) = \frac{1}{5}x - 7$

$\frac{2}{5}x + 5$

3 $\left(\frac{1}{2}x + 3\right) + (__) = \frac{5}{4}x - 1$

$\frac{3}{4}x - 4$

Subtracting a group: the minus reaches every term

$-(a+b) = -a - b$. Subtracting a whole group subtracts each term in it, so every sign in the second group flips.

WORKED EXAMPLE

Simplify:

$$\left(\frac{5}{2}x - 3\right) - \left(\frac{1}{2}x + 6\right)$$

1. Flip every sign in the second group: $\frac{5}{2}x - 3 - \frac{1}{2}x - 6$.
2. x-terms: $\frac{5}{2} - \frac{1}{2} = \frac{4}{2} = 2$.
3. Numbers: $-3 - 6 = -9$.

ANSWER $2x - 9$

WATCH FOR

Flip both signs in the second group: $-\left(\frac{7}{3}x + 4\right) = -\frac{7}{3}x - 4$.

Subtract in order, first minus second: $\frac{4}{3} - \frac{7}{3} = -1$, so $-x$.

Subtracting a group is not adding it.

YOUR TURN

Simplify:

$$\left(\frac{7}{4}y + 2\right) - \left(\frac{3}{4}y - 5\right)$$

PRACTICE · ONE CHANGE EACH TIME

1 $(6m - 1) - (2m + 3)$

whole numbers

2 $\left(\frac{2}{3}k - 4\right) - \left(\frac{5}{3}k - 4\right)$

the numbers cancel

3 $(3.5a + 2) - (1.2a - 0.8 + 0.3a)$

three terms in the group

Subtracting a group: the minus reaches every term

YOUR TURN

Simplify:

$$\left(\frac{7}{4}y + 2\right) - \left(\frac{3}{4}y - 5\right)$$

1. Flip both signs: $\frac{7}{4}y + 2 - \frac{3}{4}y + 5$.
2. y-terms: $\frac{7}{4} - \frac{3}{4} = 1$, so y. Numbers: $2 + 5 = 7$.

$y + 7$

PRACTICE

1 $(6m - 1) - (2m + 3)$

$4m - 4$

2 $\left(\frac{2}{3}k - 4\right) - \left(\frac{5}{3}k - 4\right)$

$-k$

3 $(3.5a + 2) - (1.2a - 0.8 + 0.3a)$

$2a + 2.8$

From a situation: write it whole, then simplify

Write the full expression with parentheses first, then distribute and combine. A rectangle's perimeter is $2(\text{length}) + 2(\text{width})$.

WORKED EXAMPLE

A rectangle is $(3x + 1.5)$ cm long and $\left(\frac{1}{2}x - 1\right)$ cm wide. Write and simplify its perimeter.

1. $\text{Perimeter} = 2(3x + 1.5) + 2\left(\frac{1}{2}x - 1\right)$.
2. Distribute: $6x + 3 + x - 2$.
3. Combine: $7x + 1$.

ANSWER $7x + 1$

WATCH FOR

Perimeter is all four sides: $2(\text{length}) + 2(\text{width})$, not length + width once.

Double every term: $2(x - 2) = 2x - 4$.

Keep the sign: −4, not +4.

YOUR TURN

A rectangle is $(2x - 0.5)$ ft long and $\left(\frac{3}{2}x + 4\right)$ ft wide. Write and simplify its perimeter.

PRACTICE · ONE CHANGE EACH TIME

1 A square has sides of $(x + 2.5)$ inches. Write and simplify its perimeter.

four equal sides

2 A triangle has sides of $(2x + 1)$, $(x - 3)$ and $\left(\frac{1}{2}x + 4\right)$ meters. Write and simplify its perimeter.

three different sides

3 Tickets cost $(x + 4.5)$ dollars each. Kim buys 3 tickets and uses a $5 coupon. Write and simplify her total.

not a shape

From a situation: write it whole, then simplify

YOUR TURN

A rectangle is $(2x - 0.5)$ ft long and $\left(\frac{3}{2}x + 4\right)$ ft wide. Write and simplify its perimeter.

1. $2(2x - 0.5) + 2\left(\frac{3}{2}x + 4\right) = 4x - 1 + 3x + 8.$

$7x + 7$

PRACTICE

1 A square has sides of $(x + 2.5)$ inches. Write and simplify its perimeter.

$4x + 10$

2 A triangle has sides of $(2x + 1)$, $(x - 3)$ and $\left(\frac{1}{2}x + 4\right)$ meters. Write and simplify its perimeter.

$\frac{7}{2}x + 2$

3 Tickets cost $(x + 4.5)$ dollars each. Kim buys 3 tickets and uses a $5 coupon. Write and simplify her total.

$3x + 8.5$

Distributing a fraction: multiply every term

$a(b+c) = ab + ac$. The number outside multiplies each term inside. It never adds.

WORKED EXAMPLE

Rewrite without parentheses:

$$\frac{2}{3}(9x+6)$$

1. Multiply each term by $\frac{2}{3}$: $\frac{2}{3}(9x) + \frac{2}{3}(6)$.
2. $\frac{2}{3}(9x) = 6x$, since $9 \div 3 \times 2 = 6$.
3. $\frac{2}{3}(6) = 4$.
4. Both $\frac{2}{3}(9x) + \frac{2}{3}(6)$ and $6x + 4$ are equivalent to $\frac{2}{3}(9x+6)$.

ANSWER $6x + 4$

WATCH FOR

Multiply, don't add: $\frac{5}{4}(12x)$ is $15x$, not $\left(12 + \frac{5}{4}\right)x$.

Both terms get multiplied: $15x + 25$, not $15x + 20$.

$15x + 25$ does not become $40x$: unlike terms stay apart.

YOUR TURN

Rewrite without parentheses:

$$\frac{3}{4}(8x+12)$$

PRACTICE · ONE CHANGE EACH TIME

1 $\frac{1}{2}(10x - 4)$

subtraction inside

2 $\frac{5}{2}(4x+3)$

the number doesn't come out whole

3 $-\frac{2}{5}(15x - 10)$

a negative fraction outside

Distributing a fraction: multiply every term

YOUR TURN

Rewrite without parentheses:

$\frac{3}{4}(8x + 12)$

1. $\frac{3}{4}(8x) = 6x$ and $\frac{3}{4}(12) = 9$.

$6x + 9$

PRACTICE

1 $\frac{1}{2}(10x - 4)$

$5x - 2$

---

2 $\frac{5}{2}(4x + 3)$

$10x + \frac{15}{2}$

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3 $-\frac{2}{5}(15x - 10)$

$-6x + 4$

Spot the error: multiply both terms, then stop

Two slips cause most wrong answers: multiplying only the first term, then adding an x-term to a number. An answer like $16x + 16$ is finished.

WORKED EXAMPLE

A student rewrote $\frac{3}{2}(4x + 6)$ as $12x$. What is the correct expression?

1. Correct: $\frac{3}{2}(4x) + \frac{3}{2}(6) = 6x + 9$.
2. Slip 1: the student multiplied only $4x$, getting $6x + 6$.
3. Slip 2: then added $6x$ and 6 as if they were like terms: $12x$.

ANSWER $6x + 9$

WATCH FOR

Slip 1: only the first term gets multiplied.

Slip 2: an x-term plus a number turned into one x-term.

Slip 3: the fraction was added to each term instead of multiplied.

YOUR TURN

A student rewrote $\frac{5}{3}(6x + 3)$ as $13x$. What is the correct expression?

PRACTICE · ONE CHANGE EACH TIME

1 A student rewrote $2.5(4x + 2)$ as $12x$. Correct it.

a decimal outside

2 A student rewrote $\frac{1}{2}(8x - 6)$ as $4x - 6$. Correct it.

only one slip

3 A student rewrote $\frac{3}{4}(8x + 4)$ as $\frac{35}{4}x + \frac{19}{4}$. Correct it.

a different slip: adding 3/4

Spot the error: multiply both terms, then stop

YOUR TURN

A student rewrote $\frac{5}{3}(6x + 3)$ as $13x$. What is the correct expression?

1. $\frac{5}{3}(6x) = 10x$ and $\frac{5}{3}(3) = 5$.
2. The student kept the $3(10x + 3)$, then added: $13x$.

$10x + 5$

PRACTICE

1 A student rewrote $2.5(4x + 2)$ as $12x$. Correct it.

$10x + 5$

2 A student rewrote $\frac{1}{2}(8x - 6)$ as $4x - 6$. Correct it.

$4x - 3$

3 A student rewrote $\frac{3}{4}(8x + 4)$ as $\frac{35}{4}x + \frac{19}{4}$. Correct it.

$6x + 3$

Distributing a decimal: same rule, careful multiplying

$1.5(6x - 2)$ means $1.5 \times 6x$ and $1.5 \times (-2)$. The minus stays with the 2.

WORKED EXAMPLE

Rewrite without parentheses:

$2.5(4x - 3)$

1. $2.5 \times 4x = 10x$.
2. $2.5 \times (-3) = -7.5$.

ANSWER $10x - 7.5$

WATCH FOR

Multiply, don't add: $1.5 + 6 = 7.5$ is the wrong move.

Multiply the second term too, sign and all:
$1.5 \times (-2) = -3$.

$9x - 3$ has two unlike terms; it does not shrink to $6x$.

YOUR TURN

Rewrite without parentheses:

$0.5(8x - 5)$

PRACTICE · ONE CHANGE EACH TIME

1 $1.2(5x + 10)$

addition inside

2 $0.75(4x - 8)$

hundredths outside

3 $-1.5(2x - 4)$

a negative outside: both signs flip

Distributing a decimal: same rule, careful multiplying

YOUR TURN

Rewrite without parentheses:

$0.5(8x - 5)$

1. $0.5 \times 8x = 4x$ and $0.5 \times (-5) = -2.5$.

$4x - 2.5$

PRACTICE

1 $1.2(5x + 10)$

$6x + 12$

2 $0.75(4x - 8)$

$3x - 6$

3 $-1.5(2x - 4)$

$-3x + 6$

Equivalent or not: simplify both, then compare

Two expressions are equivalent when they simplify to the same thing: the same x-coefficient and the same number. Looking alike is not enough.

WORKED EXAMPLE

Is $3(2x + 1.5) + x$ equivalent to $7x + 4.5$?

1. Simplify the first: $6x + 4.5 + x = 7x + 4.5$.
2. Compare: x-coefficient 7 and 7, number 4.5 and 4.5.
3. Quick test with $x = 0$: $3(1.5) + 0 = 4.5$ and $0 + 4.5 = 4.5$.

ANSWER Yes. Both are $7x + 4.5$.

WATCH FOR

Simplify before deciding; don't judge by looks.

"Both have an x-term and a number" does not make them equal.

Multiply both terms: $2.5(8x - 2) = 20x - 5$, not $20x - 2$.

YOUR TURN

Is $2(5x - 1) + 3$ equivalent to $10x + 2$?

PRACTICE · ONE CHANGE EACH TIME

1 Is $1.5(4x + 2) - x$ equivalent to $5x + 3$?

a term is subtracted after

2 Is $3(2x - 4) + 5$ equivalent to $6x - 4$?

subtraction inside

3 Is $-2(x - 3) + 4x$ equivalent to $2x - 6$?

a negative outside

Equivalent or not: simplify both, then compare

YOUR TURN

Is $2(5x - 1) + 3$ equivalent to $10x + 2$?

1. $10x - 2 + 3 = 10x + 1$.
2. The numbers differ: 1 and 2.

No. The first is $10x + 1$.

PRACTICE

1 Is $1.5(4x + 2) - x$ equivalent to $5x + 3$?

Yes. Both are $5x + 3$.

2 Is $3(2x - 4) + 5$ equivalent to $6x - 4$?

No. The first is $6x - 7$.

3 Is $-2(x - 3) + 4x$ equivalent to $2x - 6$?

No. The first is $2x + 6$.

A negative outside the parentheses: flip every sign inside

$-\frac{3}{4}(4x-4)$ multiplies both terms by $-\frac{3}{4}$: $-3x$ and +3. A negative times a negative is positive.

WORKED EXAMPLE

Are $-\frac{2}{3}(6x-9)+5x$ and $\frac{7}{2}x-\frac{1}{2}(5x-12)$ equivalent?

1. First: $-\frac{2}{3}(6x)=-4x$ and $-\frac{2}{3}(-9)=+6$, so $-4x+6+5x=x+6$.
2. Second: $\frac{7}{2}x-\frac{5}{2}x+6=x+6$.
3. Same x-term and same number, so they are equivalent.

ANSWER Yes. Both are $x+6$.

WATCH FOR

A negative times a negative is positive: $-\frac{3}{4}(-4)=+3$, not −3.

The same number at the end is not enough; the x-terms must match too.

YOUR TURN

Are $-\frac{1}{2}(4x-6)+3x$ and $3x-\frac{1}{2}(2x+6)$ equivalent?

PRACTICE · ONE CHANGE EACH TIME

1 Simplify:

$-\frac{3}{5}(10x-5)$

just the distributing

2 Simplify:

$\frac{9}{2}x-\frac{1}{2}(5x-8)$

the minus is between terms

3 Are $-\frac{1}{3}(6x-3)+4x$ and $\frac{5}{2}x-\frac{1}{4}(2x-4)$ equivalent?

the whole question

A negative outside the parentheses: flip every sign inside

YOUR TURN

Are $-\frac{1}{2}(4x-6)+3x$ and $3x-\frac{1}{2}(2x+6)$ equivalent?

1. First: $-2x+3+3x=x+3$.
2. Second: $3x-x-3=2x-3$.

No. $x+3$ and $2x-3$.

PRACTICE

1 Simplify:

$$-\frac{3}{5}(10x-5)$$

$-6x+3$

2 Simplify:

$$\frac{9}{2}x-\frac{1}{2}(5x-8)$$

$2x+4$

3 Are $-\frac{1}{3}(6x-3)+4x$ and $\frac{5}{2}x-\frac{1}{4}(2x-4)$ equivalent?

Yes. Both are $2x+1$.

Fee plus rate: write it, then undo in reverse order

Total = one-time fee + rate × number. The rate goes with the variable; the fee stands alone. Undo the fee first (subtract), then the rate (divide).

WORKED EXAMPLE

A bowling alley charges $4.00 for shoes plus $5.50 for each game. Lena paid $26.00. How many games did she bowl?

1. Equation: $5.5g + 4 = 26$.
2. Subtract 4: $5.5g = 22$.
3. Divide by 5.5: $g = 4$.
4. Check: $5.5(4) + 4 = 26$.

ANSWER $g = 4$

WATCH FOR

The rate goes with the variable $(12.5h)$; the fee stands alone $(+25)$.

Undo + 25 by subtracting 25, not adding it.

Check by putting the answer back into the story.

YOUR TURN

A taxi charges $3.00 plus $2.25 per mile. The ride cost $16.50. How many miles?

PRACTICE · ONE CHANGE EACH TIME

1 A gym charges $35 to join plus $18 per month. Sam has paid $125. How many months?

whole dollars

2 A plumber charges $45 for the visit plus $30 per hour. The bill was $150. How many hours?

the answer is a decimal

3 A phone costs $120 up front plus $15.50 a month. Ana has paid $213 in all. How many months?

a bigger fee

Fee plus rate: write it, then undo in reverse order

YOUR TURN

A taxi charges \$3.00 plus \$2.25 per mile. The ride cost \$16.50. How many miles?

1. $2.25m + 3 = 16.5$.
2. Subtract 3: $2.25m = 13.5$.
3. Divide: $m = 6$.

$m = 6$

PRACTICE

1 A gym charges \$35 to join plus \$18 per month. Sam has paid \$125. How many months?

$18m + 35 = 125,\ m = 5$

2 A plumber charges \$45 for the visit plus \$30 per hour. The bill was \$150. How many hours?

$30h + 45 = 150,\ h = 3.5$

3 A phone costs \$120 up front plus \$15.50 a month. Ana has paid \$213 in all. How many months?

$15.5m + 120 = 213,\ m = 6$

Equations with parentheses: undo the outside first

In $1.5(h + 1.6) = -0.6$, the whole group $(h + 1.6)$ is multiplied by 1.5. Undo that first by dividing, then undo the + 1.6.

WORKED EXAMPLE

Solve:

$$2.5(k - 1.2) = -4$$

1. Divide both sides by 2.5: $k - 1.2 = -1.6$.
2. Add 1.2: $k = -0.4$.
3. Check: $2.5(-0.4 - 1.2) = 2.5(-1.6) = -4$.
4. Or distribute first: $2.5k - 3 = -4$, so $2.5k = -1$ and $k = -0.4$.

ANSWER $k = -0.4$

WATCH FOR

Divide by the number in front; don't multiply.

Then undo + 1.6 by subtracting: $-0.4 - 1.6 = -2$.

If you distribute instead, multiply both terms: $1.5h + 2.4$.

YOUR TURN

Solve:

$$1.5(m + 2.4) = -1.5$$

PRACTICE · ONE CHANGE EACH TIME

1 $4(x - 1.5) = 10$

subtraction inside

2 $0.5(y + 3) = -2$

dividing by 0.5

3 $-2(n + 0.8) = 3$

a negative outside

Equations with parentheses: undo the outside first

YOUR TURN

Solve:

$1.5(m + 2.4) = -1.5$

1. Divide by 1.5: $m + 2.4 = -1$.
2. Subtract 2.4: $m = -3.4$.

$m = -3.4$

PRACTICE

1 $4(x - 1.5) = 10$

$x = 4$

2 $0.5(y + 3) = -2$

$y = -7$

3 $-2(n + 0.8) = 3$

$n = -2.3$

Groups: when the change happens to each one

If each of 3 students gets \$5 more, each now has $(s + 5)$, and together they have $3(s + 5)$. Writing $3s + 5$ would give out only one \$5.

WORKED EXAMPLE

Four friends each paid t dollars for a ticket plus \$2.50 for a snack. They spent \$46 in all. Find t.

1. Each friend spent $(t + 2.5)$, so $4(t + 2.5) = 46$.
2. Divide by 4: $t + 2.5 = 11.5$.
3. Subtract 2.5: $t = 9$.
4. Check: $4(9 + 2.5) = 4(11.5) = 46$.

ANSWER $t = 9$

WATCH FOR

The \$5 goes to each student: $3(s + 5)$, not $3s + 5$.

Finish the job: $s + 5 = 35$ means $s = 30$, not 35.

"Decreased" or "taken out" means subtract inside the parentheses.

YOUR TURN

Five boxes each held b books. Then 3 books were taken out of each box. Now there are 60 books. Find b.

PRACTICE · ONE CHANGE EACH TIME

1 Solve:

$$2(x + 7) = 30$$

just the equation

2 Six bags each held p pounds of rice. Then 0.5 pound was taken out of each. Now the bags hold 21 pounds in all. Find p.

taken out of each

3 Three tables each seat s people. Two more chairs are added to each table, and now 24 people can sit. Find s.

added to each

Groups: when the change happens to each one

YOUR TURN

Five boxes each held b books. Then 3 books were taken out of each box. Now there are 60 books. Find b.

1. $5(b - 3) = 60$.
2. Divide by 5: $b - 3 = 12$.
3. Add 3: $b = 15$.

$b = 15$

PRACTICE

1 Solve:

$2(x + 7) = 30$

$x = 8$

---

2 Six bags each held p pounds of rice. Then 0.5 pound was taken out of each. Now the bags hold 21 pounds in all. Find p.

$6(p - 0.5) = 21$, $p = 4$

---

3 Three tables each seat s people. Two more chairs are added to each table, and now 24 people can sit. Find s.

$3(s + 2) = 24$, $s = 6$

Fraction equations: add, then use the reciprocal

Undo − 4 by adding 4. Then $\frac{2}{3}g$ means $\frac{2}{3}$ of g: undo it by multiplying by $\frac{3}{2}$, which is the same as dividing by $\frac{2}{3}$.

WORKED EXAMPLE

Solve:

$$-2\frac{1}{4} = \frac{3}{4}x - 6$$

1. Add 6 to both sides: $-2\frac{1}{4} + 6 = 3\frac{3}{4}$, so $3\frac{3}{4} = \frac{3}{4}x$.
2. Multiply by $\frac{4}{3}$: $x = \frac{15}{4} \times \frac{4}{3} = 5$.
3. Check: $\frac{3}{4}(5) - 6 = \frac{15}{4} - \frac{24}{4} = -\frac{9}{4} = -2\frac{1}{4}$.

ANSWER $x = 5$

WATCH FOR

$-1\frac{7}{9}$ means $-\left(1 + \frac{7}{9}\right) = -\frac{16}{9}$, not $-1 + \frac{7}{9}$.

Undo $\frac{2}{3}g$ by multiplying by $\frac{3}{2}$, not by $\frac{2}{3}$.

Add 4 first: $-\frac{16}{9} + \frac{36}{9} = \frac{20}{9}$.

YOUR TURN

Solve:

$$-1\frac{1}{2} = \frac{3}{5}y - 3$$

PRACTICE · ONE CHANGE EACH TIME

1 $\frac{2}{3}a + 1 = 5$

whole numbers

2 $\frac{1}{2}b - 3 = -4\frac{1}{2}$

a negative answer

3 $-2\frac{2}{3} = \frac{4}{5}c - 6$

the answer is a mixed number

Fraction equations: add, then use the reciprocal

YOUR TURN

Solve:

$$-1\frac{1}{2} = \frac{3}{5}y - 3$$

1. Add 3: $1\frac{1}{2} = \frac{3}{5}y$.
2. Multiply by $\frac{5}{3}$: $y = \frac{3}{2} \times \frac{5}{3} = \frac{5}{2} = 2\frac{1}{2}$.

$y = 2\frac{1}{2}$

PRACTICE

1 $\frac{2}{3}a + 1 = 5$

$a = 6$

2 $\frac{1}{2}b - 3 = -4\frac{1}{2}$

$b = -3$

3 $-2\frac{2}{3} = \frac{4}{5}c - 6$

$c = 4\frac{1}{6} = \frac{25}{6}$

Two-part problems: solve, then answer the real question

Solve for the unknown first, then use it. Underline what is asked: which one, and by how much?

WORKED EXAMPLE

Pool A charges $6.50 per visit. Eli used a $10 coupon and paid $55 in all. Pool B charges $7.25 per visit and takes no coupons. For the same number of visits, which pool costs less, and by how much?

1. Pool A: $6.5v - 10 = 55$.
2. Add 10: $6.5v = 65$. Divide: $v = 10$ visits.
3. Pool B for 10 visits: $7.25 \times 10 = \$72.50$.
4. Pool A cost less, by $72.50 - 55 = \$17.50$.

ANSWER Pool A, by $17.50

WATCH FOR

Put the coupon in the equation: $24.5n - 35 = 185.5$.

Don't stop at $n = 9$: answer which gym and how much.

Compare totals, not monthly prices.

YOUR TURN

Store A sells notebooks for $3.40 each. Ty used a $5 coupon and paid $29. Store B sells them for $3.15 each, no coupons. For the same number of notebooks, which store costs less, and by how much?

PRACTICE · ONE CHANGE EACH TIME

1 Solve:

$8.5n - 12 = 64.5$

just the equation

2 A class paid $15 for a bus plus $4.50 per student. The total was $123. How many students went?

one question only

3 Plan A costs $12.50 a month with $20 off the total. Mia paid $130. Plan B costs $11 a month, nothing off. For the same months, which costs less, and by how much?

the whole problem

Two-part problems: solve, then answer the real question

YOUR TURN

Store A sells notebooks for $3.40 each. Ty used a $5 coupon and paid $29. Store B sells them for $3.15 each, no coupons. For the same number of notebooks, which store costs less, and by how much?

1. $3.4n - 5 = 29$, so $3.4n = 34$ and $n = 10$.
2. Store B: $3.15 \times 10 = \$31.50$.
3. Store A cost less, by $31.50 - 29 = \$2.50$.

Store A, by $2.50

PRACTICE

1 Solve:

$8.5n - 12 = 64.5$

$n = 9$

2 A class paid $15 for a bus plus $4.50 per student. The total was $123. How many students went?

$4.5s + 15 = 123$, $s = 24$

3 Plan A costs $12.50 a month with $20 off the total. Mia paid $130. Plan B costs $11 a month, nothing off. For the same months, which costs less, and by how much?

$m = 12$; **Plan A, by $2**

Reading a graph: the circle and the arrow

Open circle: the boundary is not included ($<$ or $>$). Closed circle: it is included ($\leq$ or $\geq$). Shading to the left means less than; to the right, greater than.

WORKED EXAMPLE

Which inequality has the solution shown?

a $4x \geq -12$ **b** $4x > -12$ **c** $x + 4 \geq -12$ **d** $4x \leq -12$



1. The graph shows $x \geq -3$: closed circle at −3, shaded right.
2. $4x \geq -12$ gives $x \geq -3$. It matches.
3. $4x > -12$ has an open circle; $x + 4 \geq -12$ has boundary −16; $4x \leq -12$ shades left.

ANSWER $4x \geq -12$

WATCH FOR

An open circle means $<$ or $>$: the number itself is not a solution.

Solve each choice to find its boundary: $x + 2.5 < 20$ gives 17.5, not 8.

Shading to the left means less than.

YOUR TURN

Which inequality has the solution shown?

a $1.5x \leq 7.5$ **b** $1.5x < 7.5$ **c** $x + 1.5 < 7.5$ **d** $1.5x > 7.5$



PRACTICE · ONE CHANGE EACH TIME

1 Write the inequality the graph shows.



open circle

2 Write the inequality the graph shows.



closed circle, negative

3 Does $0.5x \leq 3$ match the graph? Does $x - 0.5 \leq 3$?



two choices to test

Reading a graph: the circle and the arrow

YOUR TURN

Which inequality has the solution shown?

a $1.5x \leq 7.5$ **b** $1.5x < 7.5$ **c** $x + 1.5 < 7.5$

d $1.5x > 7.5$



1. The graph shows $x < 5$: open circle at 5, shaded left.
2. $1.5x < 7.5$ gives $x < 5$.

$1.5x < 7.5$

PRACTICE

1 Write the inequality the graph shows.



$x > 2$

2 Write the inequality the graph shows.



$x \leq -4$

3 Does $0.5x \leq 3$ match the graph? Does $x - 0.5 \leq 3$?



$0.5x \leq 3$ **does** $(x \leq 6)$. $x - 0.5 \leq 3$ **gives** $x \leq 3.5$**, so it does not.**

Adding or subtracting: the symbol stays put

Adding or subtracting the same number on both sides never flips the inequality. Only multiplying or dividing by a negative does that.

WORKED EXAMPLE

Solve:

$-4 + m > -7.5$

1. Add 4 to both sides: $m > -7.5 + 4$.
2. $-7.5 + 4 = -3.5$.
3. No flip: we only added.
4. Check $m = 0$: $-4 > -7.5$ is true.

ANSWER $m > -3.5$

WATCH FOR

Adding or subtracting never flips the symbol.

Undo −3 by adding 3: $-6.9 + 3 = -3.9$.

$-6.9 + 3$ is still negative: picture the number line.

YOUR TURN

Solve:

$-2 + k \leq -5.4$

PRACTICE · ONE CHANGE EACH TIME

1 $y + 1.5 \geq -2$

subtract to undo +

2 $n - 2.6 < 1.4$

add to undo −

3 $6.25 > p + 7$

the variable is on the right

Adding or subtracting: the symbol stays put

YOUR TURN

Solve:

$-2 + k \leq -5.4$

1. Add 2: $k \leq -5.4 + 2 = -3.4$.

$k \leq -3.4$

PRACTICE

1 $y + 1.5 \geq -2$

$y \geq -3.5$

---

2 $n - 2.6 < 1.4$

$n < 4$

---

3 $6.25 > p + 7$

$p < -0.75$

Dividing by a negative flips the symbol

$-2 < 3$, but multiply both by –1 and you get $2 > -3$. Multiplying or dividing by a negative reverses the order, so the symbol flips.

WORKED EXAMPLE

Solve and graph:

$$-2x \geq 7$$

1. Divide both sides by –2 and flip $\geq$ to $\leq$.
2. $x \leq -3.5$: closed circle at –3.5, shaded left.
3. Check $x = -4$: $-2(-4) = 8$, and $8 \geq 7$.

ANSWER $x \leq -3.5$



WATCH FOR

Forgetting the flip puts the shading on the wrong side.

$>$ and $<$ get open circles.

$0.5 \div (-0.5) = -1$, not 1.

YOUR TURN

Solve and graph:

$$-0.25w < 1$$

PRACTICE · ONE CHANGE EACH TIME

1 $-3p > 12$

whole numbers

2 $-1.5q \leq -6$

both sides negative

3 $0.5r < -2$

dividing by a positive

Dividing by a negative flips the symbol

YOUR TURN

Solve and graph:

$-0.25w < 1$

1. Divide by −0.25 and flip: $w > 1 \div (-0.25) = -4$.

$w > -4$: **open circle at −4, shaded right.**



PRACTICE

1 $-3p > 12$

$p < -4$: **open circle at −4, shaded left**



2 $-1.5q \leq -6$

$q \geq 4$: **closed circle at 4, shaded right**



3 $0.5r < -2$

$r < -4$: **open circle at −4, shaded left. No flip.**



Words to an inequality: product, sum, at least, at most

Translate piece by piece. "The product of x and –0.5" is $-0.5x$. "At least" means $\geq$ and "at most" means $\leq$.

WORKED EXAMPLE

The product of a number x and –4 is at least 10. What are the possible values of x?

1. Write it: $-4x \geq 10$.
2. Divide by –4 and flip: $x \leq -2.5$.
3. Check $x = -3$: $-4(-3) = 12$, and 12 is at least 10.

ANSWER $x \leq -2.5$

WATCH FOR

Divide to undo multiplying: $x > 2.5$ came from multiplying –5 by –0.5.

Dividing by a negative flips the symbol: $x < 10$ forgot to.

"At least" includes the number: $\geq$.

YOUR TURN

The product of a number y and –0.5 is greater than 3. What are the possible values of y?

PRACTICE · ONE CHANGE EACH TIME

1 A number n divided by 3 is at most 4.

"at most", dividing

2 The sum of a number x and 2.5 is less than –1.

a sum

3 The product of –2 and a number z is at most –9.

flip again

Words to an inequality: product, sum, at least, at most

YOUR TURN

The product of a number y and −0.5 is greater than 3. What are the possible values of y?

1. $-0.5y > 3$. Divide by −0.5 and flip: $y < -6$.

$y < -6$

PRACTICE

1 A number n divided by 3 is at most 4.

$\frac{n}{3} \leq 4$, **so** $n \leq 12$

2 The sum of a number x and 2.5 is less than −1.

$x + 2.5 < -1$, **so** $x < -3.5$

3 The product of −2 and a number z is at most −9.

$-2z \leq -9$, **so** $z \geq 4.5$

Matching a graph: solve every choice, check three things

Solve each choice, then compare three things with the graph: the boundary number, open or closed, and the direction.

WORKED EXAMPLE

The graph shows the solution of one choice. Which?

a $x + 2 \leq 5$ **b** $-2x \leq -6$ **c** $\frac{x}{3} < 1$ **d** $x - 1 \leq 3$



1. The graph: closed circle at 3, shaded left, so $x \leq 3$.
2. **a** $x \leq 3$: matches.
3. **b** $x \geq 3$: wrong direction. **c** $x < 3$: open circle.
   d $x \leq 4$: wrong boundary.

ANSWER **a** $x + 2 \leq 5$

WATCH FOR

Check all three: boundary, circle, direction.

Don't stop at the first choice with the right number.

YOUR TURN

Which matches the graph?

a $-3x < 3$ **b** $x + 4 \geq 3$ **c** $\frac{x}{2} > -1$ **d** $-x > 1$



PRACTICE · ONE CHANGE EACH TIME

1 Solve $-0.5x \geq 2$. Give the boundary, open or closed, and direction.

divide by a negative

2 Solve $x - 3.5 > -1.5$. Boundary, open or closed, direction?

add

3 Solve $\frac{x}{-2} < 1.5$. Boundary, open or closed, direction?

multiply by a negative

Matching a graph: solve every choice, check three things

YOUR TURN

Which matches the graph?

a $-3x < 3$ **b** $x + 4 \geq 3$ **c** $\frac{x}{2} > -1$ **d** $-x > 1$



1. The graph: open circle at −1, shaded right, so $x > -1$.
2. **a** $x > -1$: matches.
3. **b** $x \geq -1$: closed circle. **c** $x > -2$: wrong boundary. **d** $x < -1$: wrong direction.

a $-3x < 3$

PRACTICE

1 Solve $-0.5x \geq 2$. Give the boundary, open or closed, and direction.

$x \leq -4$: **−4, closed, left**



2 Solve $x - 3.5 > -1.5$. Boundary, open or closed, direction?

$x > 2$: **2, open, right**



3 Solve $\frac{x}{-2} < 1.5$. Boundary, open or closed, direction?

$x > -3$: **−3, open, right**

